LM05 Portfolio Mathematics

Version 1.0

1. Introduction

This learning module covers:

- Calculating the expected return and variance of a portfolio
- Calculating covariance and correlation of portfolio returns using a joint probability function
- Portfolio risk measures: Roy's safety-first ratio

2. Portfolio Expected Return and Variance of Return

Expected Return

A portfolio's expected return can be calculated as:

$$E(R_P) = w_1E(R_1) + w_2E(R_2) + \ldots + w_nE(R_n)$$

where:
w_n = portfolio weight of n^{th} security in the portfolio
R_n = expected return of n^{th} security in the portfolio
n = number of securities in the portfolio

We will discuss portfolio expected return and variance of return using a two-stock portfolio.

Example

40% of the portfolio is invested in Stock A and 60% is invested in Stock B. As shown in the table below, the expected return of each stock depends on the economic scenario.

Scenario	P(Scenario)	Expected returns of A	Expected returns of B
Recession	0.25	2%	4%
Normal	0.50	8%	10%
Boom	0.25	12%	16%

Calculate the expected return of A and B.

Solution:

Given the data presented above:

The expected return of A is: 0.25 x 2 + 0.50 x 8 + 0.25 x 12 = 7.5%.

The expected return of B is: 0.25 x 4 + 0.50 x 10 + 0.25 x 16 = 10%.

Expected return of the portfolio = weight of A in the portfolio x expected return of A + weight of stock B in the portfolio x expected return of B = 0.4 x 7.5 + 0.6 x 10 = 9%

The expected portfolio return is 9%. As the term implies, this is the expected return. The actual return will vary around 9%. The amount of variability is measured by the variance. In order to determine the variance of return, we must first calculate the covariance.

Covariance

Covariance tells us how movements in a random variable vary with movements in another random variable, whereas variance tells us how a random variable varies with itself. Assume there are two random variables R_i and R_j. The covariance between R_i and R_j (used to measure how they move together) is given by:

$$\text{Cov}\left(R_i, R_j\right) = E\left([R_i - ER_i][R_j - ER_j]\right)$$

where:
ER_i = expected return for variable R_i
ER_j = expected return for variable R_j

Example

Continuing with our previous example, calculate the covariance of returns between A and B.

Solution:

Say R_i represents the return on A and R_j represents the return on B, we have already calculated the expected returns of A and B as 7.5% and 10% respectively. The covariance of returns is:

$E\left[(Ri - 7.5)\ (Rj - 10)\right]$

= 0.25(2% - 7.5%)(4% - 10%) + 0.5(8% - 7.5%)(10% - 10%) + 0.25(12% - 7.5%)(16% - 10%)

= 0.000825 + 0 + 0.000675 = 0.0015

We can interpret the sign of covariance as:

- Covariance of returns is negative if, when the return on one asset is above its expected value, the return on the other asset tends to be below its expected value.
- Covariance of returns is 0 if the returns on the assets are unrelated.
- Covariance of returns is positive when the returns on both assets tend to be on the same side (above or below) their expected values at the same time.

Correlation

The problem with covariance is that it can vary from negative infinity to positive infinity which makes it difficult to interpret. To address this problem, we use another measure called correlation. Correlation is a standardized measure of the linear relationship between two variables with values ranging between -1 and +1.

- A correlation of 0 (uncorrelated variables) indicates an absence of any linear (straight-line) relationship between the variables.
- A correlation of +1 indicates a perfect positive relationship.
- A correlation of -1 indicates a perfect negative relationship.

It is computed as:

$$\rho(R_i, R_j) = Cov(R_i, R_j)/\sigma(R_i)\,\sigma(R_j)$$

We will now apply this formula to calculate the correlation between the returns of A and B from our example. We have already shown that the covariance of returns is 0.0015. In order to calculate the correlation, we need the standard deviation of A and B. Using a financial calculator, we can determine that the standard deviation of A is 0.0357 and the standard deviation of B is 0.0424. The correlation, $\rho(A, B) = \frac{Cov(A,B)}{\sigma(A)\sigma(B)}$ = 0.0015/(0.0357 x 0.0424) = 0.99.

Instructor's Note: The keystrokes for calculating the standard deviation of A are shown below.

Keystrokes	***Explanation***	***Display***
[2nd] [DATA]	Enters data entry mode	
[2nd] [CLR WRK]	Clears data register	X01
0.02 [ENTER]	1st possible value of random variable	X01 = 0.02
[↓] 25 [ENTER]	Probability of 25% for X01	Y01 = 25
[↓] 0.08 [ENTER]	2nd possible value of random variable	X02 = 0.08
[↓] 50 [ENTER]	Probability of 50% for X02	Y02 = 50
[↓] 0.12 [ENTER]	3rd possible value of random variable	X03 = 0.12
[↓] 25 [ENTER]	Probability of 25% for X03	Y03 = 25
[2nd] [STAT]	Puts calculator into stats mode	
[2nd] [SET]	Press repeatedly till you see →	1-V
[↓]	Total number of entries	N = 100
[↓]	Expected value of random variable	X = 0.075
[↓]	Sample standard deviation	Sx = 0.0359
[↓]	Population standard deviation	σx = 0.0357

The correlation of 0.99 (almost 1) implies a very strong positive relationship between the returns of A and B. This is more meaningful than the covariance number of 0.0015 which tells us that there is a positive relationship between the returns of A and B but does not give a sense for the strength of the relationship.

Variance of returns

Once we know the covariance, we can calculate the variance of a portfolio using this formula:

$$\sigma^2(R_P) = w_1^2\sigma_1^2\,(R_1) + w_2^2\sigma_2^2\,(R_2) + 2w_1w_2Cov\,(R_1R_2)$$

Example

Continuing with our example, the variance of the portfolio is

- Weight of the first asset, w_1 = 0.40
- Weight of the second asset, w_2 = 0.60
- Standard deviation of first asset = 0.0357
- Standard deviation of second asset = 0.0424
- Covariance between the two assets = 0.0015

Variance of the portfolio = $0.4^2 \times 0.0357^2 + 0.6^2 \times 0.0424^2 + 2 \times 0.4 \times 0.6 \times 0.0015 = 0.00157$

Standard deviation of the portfolio = $\sqrt{0.00157} = 0.0396$

The portfolio variance for a three-asset portfolio can be calculated as:

$$\sigma^2(R_P) = w_1^2\sigma_1^2\,(R_1) + w_2^2\sigma_2^2\,(R_2) + w_3^2\sigma_3^2\,(R_3) + 2w_1w_2\mathrm{Cov}\,(R_1R_2) + 2w_1w_3\mathrm{Cov}\,(R_1R_3) + 2w_2w_3\mathrm{Cov}\,(R_2R_3)$$

Sometimes we may be provided a variance-covariance matrix. We can use this matrix to calculate the portfolio standard deviation as illustrated in the following example:

Example:
(*This is based on Question 9 from the curriculum.*)

An analyst has designed the following three asset portfolio:

	Asset 1	**Asset 2**	**Asset 3**
Expected return	5%	6%	7%
Portfolio weight	0.20	0.30	0.50

Variance-Covariance Matrix

	Asset 1	**Asset 2**	**Asset 3**
Asset 1	196	105	140
Asset 2	105	225	150
Asset 3	140	150	400

Instructor's Note: A variance covariance matrix is defined as a square matrix where the diagonal elements represent the variance and off-diagonal elements represent the covariance. In the above matrix, the variance of Asset 1, Asset 2 and Asset 3 are 196, 225 and 400 respectively. The covariance between Asset 1 and Asset 2 is 105. The covariance between Asset 1 and Asset 3 is 140. The covariance between Asset 2 and Asset 3 is 150.

Calculate the expected return and standard deviation of the portfolio.

Solution:

Expected return = $w_1E(R_1) + w_2E(R_2) + w_3E(R_3)$ = 0.20(5%) + 0.30 (6%) + 0.50(7%) = 6.3%

Portfolio variance = $w_1^2\sigma_1^2\ (R_1) + w_2^2\sigma_2^2\ (R_2) + w_3^2\sigma_3^2\ (R_3) + 2w_1w_2\text{Cov}\ (R_1R_2) + 2w_1w_3\text{Cov}\ (R_1R_3) + 2w_2w_3\text{Cov}\ (R_2R_3)$

= $(0.20)^2(196) + (0.30)^2(225) + (0.50)^2(400) + 2(0.20)(0.30)(105) + 2(0.20)(0.50)(140) + 2(0.30)(0.50)(150) = 213.69$

Standard deviation is the square root of variance. Therefore, portfolio standard deviation = sqrt (213.69) = 14.62%

A variance covariance matrix can also be used to calculate correlation as illustrated in the following example.

Example:
(*This is based on Question 4 from the curriculum*.)

An analyst develops the following covariance matrix of returns.

	Hedge Fund	**Market Index**
Hedge Fund	256	110
Market Index	110	81

Calculate the correlation of returns between the hedge fund and the market index.

Solution:

$\rho(R_i, R_j) = \text{Cov}(R_i, R_j)/\sigma(R_i)\ \sigma(R_j)$

$$\rho(R_i, R_j) = \frac{110}{\sqrt{256} \times \sqrt{81}} = \frac{110}{16 \times 9} = 0.764$$

3. Forecasting Correlation of Returns: Covariance Given a Joint Probability Function

The same information we saw in section 2 can also be presented in the form of a joint probability function.

Example:

Scenario	**P(Scenario)**	**Expected returns of A**	**Expected returns of B**
Recession	0.25	2%	4%
Normal	0.50	8%	10%
Boom	0.25	12%	16%

This information can also be presented as a joint probability function of A's and B's returns:

	R_B = 4%	**R_B = 10%**	**R_B = 16%**
R_A = 2%	0.25	0	0
R_A = 8%	0	0.50	0
R_A = 12%	0	0	0.25

Row 1 and Column 1 represent the returns of A and B respectively. The other cells contain probabilities.

We can calculate the covariance in the same way that we did in section 2.

The expected return of A is: 0.25 x 2 + 0.50 x 8 + 0.25 x 12 = 7.5%.

The expected return of B is: 0.25 x 4 + 0.50 x 10 + 0.25 x 16 = 10%.

Covariance of returns = E [(Ri – 7.5) (Rj – 10)]

= 0.25(2% - 7.5%) (4% - 10%) + 0.5(8% - 7.5%) (10% - 10%) + 0.25(12% - 7.5%) (16% - 10%)

= 0.000825 + 0 + 0.000675 = 0.0015

Instructor's Note: Do the above calculations by only looking at the data in the joint probability table. On the exam you may be presented information in either format (normal table or joint probability table).

4. Portfolio Risk Measures: Applications of The Normal Distribution

Shortfall risk

Shortfall risk is the risk that portfolio's return will fall below a specified minimum level of return over a given period of time.

Safety first ratio

Safety first ratio is used to measure shortfall risk. It is calculated as:

$$\text{SF Ratio} = \frac{[R_P - R_L]}{\sigma_P}$$

where:
R_p = Expected portfolio return
R_L = Threshold level
σp = Standard deviation of portfolio returns

The portfolio with the highest SF-Ratio is preferred, as it has the lowest probability of falling below the target return.

Roy's safety-first criteria

It states that an optimal portfolio minimizes the probability that the actual portfolio return will fall below the target return.

Example

An investor is considering two portfolios A and B. Portfolio A has an expected return of 10% and a standard deviation of 2%. Portfolio B has an expected return of 15% and a standard deviation of 10%. The minimum acceptable return for the investor is 8%. According to Roy's safety-first criteria, which portfolio should the investor select?

Solution:

$$SF_A = \frac{10 - 8}{2} = 1$$

$$SF_B = \frac{15 - 8}{10} = 0.7$$

Since A has a higher safety-first ratio, the investor should select portfolio A.

If the risk-free rate is set as the threshold level R_L, the safety-first ratio becomes the Sharpe ratio.

$$\text{Sharpe Ratio} = \frac{[R_P - R_f]}{\sigma_P}$$

Summary

LO: Calculate and interpret the expected value, variance, standard deviation, covariances, and correlations of portfolio returns.

Expected return: $E(R_P) = w_1E(R_1) + w_2E(R_2) + ... + w_nE(R_n)$

Covariance: $\text{Cov}\left(R_i, R_j\right) = E\left([R_i - ER_i][R_j - ER_j]\right)$

Correlation: $\rho(R_i, R_j) = \text{Cov}(R_i, R_j)/\sigma(R_i)\,\sigma(R_j)$

Variance of a 2-asset portfolio: $\sigma^2(R_P) = w_1^2\sigma_1^2\,(R_1) + w_2^2\sigma_2^2\,(R_2) + 2w_1w_2\text{Cov}\,(R_1R_2)$

Variance of a 3-asset portfolio: $\sigma^2(R_P) = w_1^2\sigma_1^2\,(R_1) + w_2^2\sigma_2^2\,(R_2) + w_3^2\sigma_3^2\,(R_3) + 2w_1w_2\text{Cov}\,(R_1R_2) + 2w_1w_3\text{Cov}\,(R_1R_3) + 2w_2w_3\text{Cov}\,(R_2R_3)$

LO: Calculate and interpret the covariance and correlation of portfolio returns using a joint probability function for returns.

The covariance of portfolio returns can also be estimated using a joint probability function. This value is derived by summing all possible deviation cross-products weighted by the appropriate joint probability.

LO: Define shortfall risk, calculate the safety-first ratio, and identify an optimal portfolio using Roy's safety-first criterion.

Shortfall risk is the risk that portfolio's return will fall below a specified minimum level of return over a given period of time.

Safety first ratio is used to measure shortfall risk. It is calculated as:

$$\text{SF Ratio} = \frac{[R_P - R_L]}{\sigma_P}$$

According to Roy's safety-first criterion, the portfolio with the highest SF-Ratio is preferred, as it has the lowest probability of falling below the target return.